# Project Progress Report

Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

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## 1 Project Overview

This project provides a systematic numerical characterization of hyperuniform point patterns in one dimension. Two complementary metrics are computed for each pattern:

- **Hyperuniformity exponent** $\alpha$: characterizes how $S(k) \sim |k|^\alpha$ as $k \to 0$.
- **Surface-area coefficient** $\bar{\Lambda}$: measures residual disorder within the strongest (Class I) patterns; $\bar{\Lambda} = 1/6$ is the global minimum (integer lattice).

Three universality classes arise from the large-$R$ behavior of the number variance $\sigma^2(R)$:

| Class | Condition | Variance | $\bar{\Lambda}$ |
|---|---|---|---|
| I | $\alpha > 1$ | bounded | finite |
| II | $\alpha = 1$ | $\sim C_{II} \ln R$ | diverges |
| III | $0 < \alpha < 1$ | $\sim A_{III} R^{1-\alpha}$ | diverges |

The project is organized by "weeks," each corresponding to a checkpoint meeting with Prof. Torquato (not necessarily one calendar week). Three such meetings have taken place to date, yielding 27 characterized patterns and a Junior Paper (§1–§3 complete, 9 pages; results sections pending).

## 2 Literature Review

This section surveys the primary literature that motivates, informs, and contextualizes the project. Twelve papers are reviewed, organized thematically.

### 2.1 Foundations of Hyperuniformity

**Torquato & Stillinger (2003).** The foundational paper defines **hyperuniformity** as the condition $S(\mathbf{k} \to \mathbf{0}) = 0$, where $S(\mathbf{k})$ is the structure factor. For a $d$-dimensional point pattern the **number variance** $\sigma^2(R) \equiv \langle N(R)^2 \rangle - \langle N(R) \rangle^2$ grows at most as the surface area $R^{d-1}$ for large $R$, in contrast to the $R^d$ growth of a Poisson process. Three universality classes are introduced: Class I ($\alpha > 1$, bounded $\bar{\Lambda}$), Class II ($\alpha = 1$, $\sigma^2 \sim C_{II} \ln R$), Class III ($0 < \alpha < 1$, $\sigma^2 \sim A_{III} R^{1-\alpha}$).

The **surface-area coefficient**

$$\bar{\Lambda} = \lim_{R \to \infty} \frac{\sigma^2(R)}{2\rho R^{d-1} A_1(R)}$$

(with $A_1(R)$ the window surface area) serves as an order metric for Class I patterns. For $d = 1$ the integer lattice achieves $\bar{\Lambda}_{\min} = 1/6$, derived from the exact formula $\sigma^2(R) = \{R\}(1 - 2\{R\})$ where $\{R\}$ is the fractional part of $R$. In higher dimensions, Bravais lattices minimize $\bar{\Lambda}$ among periodic patterns. The paper establishes the conceptual framework adopted throughout this project.

**Zachary & Torquato (2009).** This paper extends hyperuniformity to **two-phase heterogeneous media**. In this setting the relevant quantity is the **spectral density** $\tilde{\chi}_V(\mathbf{k})$ — the Fourier transform of the two-point correlation of the indicator function $I(\mathbf{x})$ for phase 2. Hyperuniformity now requires $\tilde{\chi}_V(\mathbf{k} \to \mathbf{0}) = 0$, in direct analogy with $S(\mathbf{k} \to \mathbf{0}) = 0$ for point patterns.

A central technical contribution is the **theta-function (Bragg-sum) method** for computing $\bar{\Lambda}$ analytically from the Fourier-space structure of the pattern: Bragg peaks at reciprocal-lattice vectors $\mathbf{k}_j$ contribute to $\sigma^2(R)$ through overlap integrals of the window function. This method is applied to tabulate $\bar{\Lambda}$ for $d = 1, 2, 3$ Bravais lattices and one-dimensional quasicrystals. The Fibonacci chain result $\bar{\Lambda} = 0.20110$ (Table 1 of that paper) serves as the primary literature benchmark for this project's numerical validation.

### 2.2 Substitution Tilings: Classification

**Öguz, Socolar, Steinhardt, Torquato (2019).** This paper provides a comprehensive classification of hyperuniformity and **anti-hyperuniformity** in all one-dimensional substitution tilings. Given a substitution matrix $\mathbf{M}$ with Perron–Frobenius eigenvalue $\lambda_1$ and second-largest eigenvalue $\lambda_2$, the **eigenvalue formula**

$$\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln \lambda_1}$$

expresses the hyperuniformity exponent entirely in terms of the matrix spectrum. The exponent $\alpha$ fills $[-1, 3]$ densely over the space of all valid substitution matrices, including anti-hyperuniform cases ($\alpha < 0$) where the structure factor *diverges* at the origin.

The Pisot–Vijayaraghavan (PV) property — $|\lambda_i| < 1$ for all $i \geq 2$ — is shown to be necessary and sufficient for the existence of **Bragg peaks** (non-zero weight in the diffraction measure at isolated $k$-values). Non-Pisot substitutions lack Bragg peaks entirely, and their diffraction measure is purely continuous; $Z(k)$ then decays faster than any power law, placing them outside the $\alpha$-classification scheme. Concrete results: Fibonacci $\alpha = 3$ (Class I), Period-Doubling $\alpha = 1$ (Class II). These two cases are used throughout this project.

**Bombieri & Taylor (1986).** Although no PDF is available, this paper is a key reference: the authors studied the *cubic* 3-letter substitution $a \to aac$, $b \to ac$, $c \to b$ (equivalently encoded by a $3 \times 3$ integer matrix) and showed that it produces a well-defined diffraction pattern with Bragg peaks. With $\lambda_1 \approx 2.247$ and $|\lambda_2| \approx 0.802$ the eigenvalue formula gives $\alpha = 1.545$, the **first known example filling the gap** $1 < \alpha < 2$. This chain was reproduced in the present project with $\bar{\Lambda} = 0.377$.

### 2.3 Quasicrystals and Projection Methods

**Shechtman, Blech, Gratias, Cahn (1984).** The experimental discovery of an Al-Mn alloy with icosahedral diffraction symmetry — forbidden by classical crystallography — launched the field of quasicrystals. Bragg peaks appear at irrational wavevector ratios, implying long-range order without periodicity. This motivates the theoretical framework of *cut-and-project* (projection) methods in which a higher-dimensional periodic lattice is projected onto a lower-dimensional physical space along an irrational direction.

**Öguz, Macé, Torquato (2017).** This paper analyzes hyperuniformity of projection-method quasicrystals analytically. For the ideal Fibonacci strip (projection window aligned with the golden ratio), the structure factor satisfies $Z(k) \sim k^4$, giving $\alpha = 3$ (Class I). A nonideal strip (irrational window boundary) degrades to $\alpha = 1$ (logarithmic variance, Class II): the rational approximants to the boundary introduce quasi-periodic modulations that slow the decay of $S(k)$.

The paper derives an **analytical upper bound** $\sigma^2(R) \leq 1/4$ for the Fibonacci chain, consistent with the numerical Fibonacci $\bar{\Lambda} = 0.201 < 1/4$ found in this project (and with $\bar{\Lambda}_{\text{Silver}} = 1/4$ as a candidate upper-bound saturation for $n = 2$). The two-point correlation $g_2(r)$ is shown to be piecewise quadratic with inflection points at the Fibonacci spacings 1 and $\tau = (1 + \sqrt{5})/2$.

### 2.4 Stealthy Hyperuniform Ground States

**Batten, Stillinger, Torquato (2008).** This paper introduced the concept of **collective-coordinate optimization**: given a target function $\Phi = \sum_{|\mathbf{k}| \leq K} S(\mathbf{k})$, gradient descent on particle positions drives $S(\mathbf{k}) \to 0$ for $|k| \leq K$, engineering a **stealthy** configuration. The resulting patterns are highly degenerate ground states (many configurations satisfy the stealth condition) and are generally *disordered*. Special cases include super-ideal gas variants and equi-luminous structures. A key insight: stealth can be achieved with particles that appear to have no macroscopic order, yet exhibit complete isotropy and hidden long-range correlations. This paper predates the thermodynamic ensemble framework of Torquato et al. (2015) and provides the algorithmic foundation for all stealthy ensemble generation.

**Torquato, Zhang, Stillinger (2015) — *Physical Review X*.** This paper develops the **exact ensemble theory** for stealthy hyperuniform disordered ground states. The control parameter is

$$\chi = \frac{M(K)}{d(N-1)},$$

where $M(K)$ is the number of independently constrained wavevectors within a ball of radius $K$, $N$ is the particle number, and $d$ the dimension. For $\chi < 1/2$ the ground-state manifold is *disordered* (positive entropy); for $\chi > 1/2$ only periodic configurations satisfy the constraints. All thermodynamic quantities — isothermal compressibility $\kappa_T$, pair correlations $g_2(r)$, structure factor $S(k)$ — are derived analytically in the ensemble limit. The paper also derives the relation $K^* = 4\pi\rho\chi$ (for $d = 1$) between the exclusion radius $K^*$ and the control parameter $\chi$.

In $d = 1$ the paper's Table V provides reference values of $\bar{\Lambda}$ for stealthy configurations, which are well approximated by $\bar{\Lambda} \approx 1/(\pi^2\chi)$ — a formula derived independently in this project and validated against the collaborator's ensemble data.

### 2.5 Perturbed Lattices and Cloaking

**Klatt, Kim, Torquato (2020).** This paper introduces and analyzes the **uncorrelated random lattice (URL)**: each site of the integer lattice $\mathbb{Z}^d$ is displaced independently and uniformly within a hypercube $[-a/2, a/2]^d$. In one dimension the **exact formula**

$$\bar{\Lambda}(a) = \frac{1 + a^2}{6}$$

is derived analytically (for $a \leq 1$) and extended via periodicity to all $a > 0$. Crucially, the URL always has $\alpha = 2$ regardless of $a$: the uniform displacement smears the lattice Bragg peaks into a continuous background, but the origin $k = 0$ retains the same power-law suppression $S(k) \sim k^2$.

The central discovery is **cloaking**: when the displacement amplitude $a$ equals an integer (1, 2, 3, ...), the form factor $\tilde{f}(k) = \sin(\pi k a)/(\pi k a)$ vanishes at every non-zero reciprocal lattice vector, completely eliminating the Bragg peaks from the diffraction pattern. At $a = 1$: $\bar{\Lambda} = 1/3$, which this project identifies as the *exact limit* of the metallic-mean sequence $\bar{\Lambda}(\mu_n) \to 1/3$ as $n \to \infty$. The $\tau$ order metric (based on pair correlations) diverges between cloaking events, providing a complementary characterization of the structural transition.

### 2.6 Diffusion Spreadability

**Torquato (2021) — *Physical Review E.*** This paper introduces **diffusion spreadability** $\mathcal{S}(t)$ as a dynamic probe of microstructure. Consider a two-phase medium (phases 1 and 2) with a solute initially confined to phase 1; $\mathcal{S}(t)$ measures the fraction of solute transferred to phase 2 by time $t$. For hyperuniform media with exponent $\alpha$ the long-time behavior is

$$E(t) \equiv \mathcal{S}(\infty) - \mathcal{S}(t) \ \sim \ t^{-(d+\alpha)/2},$$

while stealthy media show *exponential* decay $E(t) \sim e^{-ct}$. At short times $\mathcal{S}(t) \sim t^{1/2}$ in all cases, providing access to the **specific surface** $s$ (surface area per unit volume of the interface). Applications include NMR/MRI characterization of porous media, where the diffusion signal encodes structural information across length scales. The 1D formula $E(t) \sim t^{-(1+\alpha)/2}$ is used in this project to extract $\alpha$ via a log-log regression of the excess spreadability.

**Wang & Torquato (2022) — *Physical Review Applied.*** This paper applies spreadability to a broad set of 2D and 3D model microstructures: URL packings, stealthy ground states, Debye random media (nonhyperuniform), Bravais lattices, and real porous media (Fontainebleau sandstone). A practical algorithm is presented to extract $\alpha$ from the slope of $\ln E(t)$ versus $\ln t$ in the asymptotic regime. For one-dimensional structures the paper gives $\phi_2 = 0.35$ for the Fibonacci chain (volume fraction of phase 2 in the two-phase embedding), which is used throughout this project for all 1D spreadability calculations.

**Hitin-Bialus, Maher, Steinhardt, Torquato (2024).** This paper achieves high-precision extraction of $\alpha$ for quasiperiodic tilings via spreadability. By exploiting the **self-similarity** of substitution tilings, the small-$k$ Bragg-peak regime (where oscillations dominate) can be truncated, leaving a clean power-law tail in $E(t)$. Results: Fibonacci $\alpha = 3.0000 \pm 0.000008$; Period-Doubling $\alpha = 1.0000 \pm 0.00001$; Penrose tiling (2D) $\alpha = 5.97 \pm 0.06$ (close to the Class I value $d + 1 = 3$ for a 2D tiling with the Pisot property). The paper confirms $\phi_2 = 0.35$ for all 1D chains studied and provides the most precise numerical validation of the eigenvalue formula (Öguz et al. 2019) to date.

### 2.7 Order Metrics and Comparative Rankings

**Middlemas, Stillinger, Torquato (2019).** This paper computes $\bar{\Lambda}$ for all Barlow sphere-packing sequences in $d = 3$, parametrized by fcc/hcp stacking codes. A linear model $\bar{\Lambda}(\alpha) = \alpha\bar{\Lambda}_{\text{fcc}} + (1-\alpha)\bar{\Lambda}_{\text{hcp}}$ holds for periodic stacking sequences (where $\alpha$ is the fraction of fcc layers). The dominant factor in $\bar{\Lambda}$ is the local cluster geometry: fcc-like environments (60° tetrahedral) are more hyperuniform than hcp-like environments (90° tetrahedral). Theta-series methods are used for all Bragg-peak sums, establishing the Zachary–Torquato (2009) technique as standard. The paper demonstrates that $\bar{\Lambda}$ serves as a sensitive discriminator among structures that share the same pair-correlation function to short range.

**Morse, Steinhardt, Torquato (2024).** This paper studies ordered versus disordered stealthy hyperuniform systems across spatial dimensions $d = 1$–$4$. In $d = 1$ the densest Fourier-space hard-sphere packing — which saturates the stealthy constraint with the fewest degrees of freedom — corresponds exactly to a Bravais lattice. In higher dimensions, disordered stealthy configurations can achieve *lower* packing density than their crystalline counterparts, counter to the usual expectation. The **order metric** $\tau$ (based on pair-distance statistics) is used as a complement to $\bar{\Lambda}$: $\tau$ is large for ordered (lattice-like) stealthy configurations and small for disordered ones. The results motivate using both $\bar{\Lambda}$ and $\tau$ to fully characterize stealthy hyperuniform materials, and demonstrate that stealth alone does not determine the degree of order.

### 2.8 Published Reference Values

The table below collects all $\alpha$ and $\bar{\Lambda}$ values that appear explicitly in the published literature, restricted to the $d = 1$ patterns directly relevant to this project plus the 2D Penrose tiling. Values marked † were confirmed independently by this project.

| **Pattern** | **Class** | $\alpha$ **(published)** | $\bar{\Lambda}$ **(published)** | **Source** |
|---|---|---|---|---|
| Integer lattice ($d = 1$) | I | $\infty$ | 1/6 † | Torquato & Stillinger (2003) |
| Fibonacci chain | I | 3 (exact) † | 0.20110 † | Zachary & Torquato (2009); Öguz et al. (2019); Hitin-Bialus et al. |
| Period-Doubling chain | II | 1 (exact) † | diverges | Öguz et al. (2019); Hitin-Bialus et al. |
| Bombieri-Taylor chain | I | 1.545 † | — | Bombieri & Taylor (1986); Öguz et al |
| URL ($a > 0$, $d = 1$) | I | 2 † | $(1 + a^2)/6$ † | Klatt, Kim, Torquato (2020) |
| URL cloaked ($a = 1$) | I | 2 † | 1/3 † | Klatt, Kim, Torquato (2020) |
| Stealthy (d=1, various $\chi$) | I | n/a † | see PRX Table V | Torquato, Zhang, Stillinger (2015) |
| Penrose tiling ($d = 2$) | I | $5.97 \pm 0.06$ | — | Hitin-Bialus et al. (2024) |

† Independently confirmed by numerical computation in this project. "—" indicates the value was not reported in the cited work. $\bar{\Lambda}$ values for the Period-Doubling and Bombieri-Taylor chains were *not* published prior to this project. The Penrose tiling $\alpha$ is consistent with $d + \alpha_{1\mathrm{D}} = 2 + 3 = 5$ but the precise interpretation in 2D remains open.

## 3 Week 1

### 3.1 Code Benchmarking and Validation

The core numerical algorithm for computing $\sigma^2(R)$ was validated against the 1D Poisson process ($\sigma^2 = 2\rho R$) to within **1.1% mean relative error** across the full $R$-range. Output: `fig1_poisson_benchmark.png`.

### 3.2 Quasicrystal Generation

Two generation methods were implemented:

1. **Substitution method** (`substitution_tilings.py`): iterative rule application up to $N \sim 10^7$ for all five metallic-mean chains.
2. **Cut-and-project method** (`projection_method.py`): Fibonacci chain from $\mathbb{Z}^2$ projection. Agreement with substitution: $< 0.1\%$ on $\bar{\Lambda}$.

| **Chain** | **Metallic mean** $\mu_n$ | $\lvert\det M\rvert$ | $\alpha$ (exact) |
|---|---|---|---|
| Fibonacci ($n = 1$) | $(1 + \sqrt{5})/2 \approx 1.618$ | 1 | 3 |
| Silver ($n = 2$) | $1 + \sqrt{2} \approx 2.414$ | 1 | 3 |
| Bronze ($n = 3$) | $(3 + \sqrt{13})/2 \approx 3.303$ | 1 | 3 |
| Copper ($n = 4$) | $2 + \sqrt{5} \approx 4.236$ | 1 | 3 |
| Nickel ($n = 5$) | $(5 + \sqrt{29})/2 \approx 5.193$ | 1 | 3 |

All $2 \times 2$ metallic-mean matrices have $\lvert\det M\rvert = 1$, which forces $\lvert\lambda_2\rvert = 1/\lambda_1$ and thus $\alpha = 3$ exactly for all of them.

### 3.3 Number Variance and $\bar{\Lambda}$

$\sigma^2(R)$ was computed for all five metallic-mean chains at $N \sim 10^6$–$10^7$. All are Class I (bounded, oscillating variance). The exact formula for the integer lattice is $\sigma^2(R) = \{R\}(1 - 2\{R\})$ where $\{R\}$ denotes the fractional part.

| **Pattern** | $\bar{\Lambda}$ | **Notes** |
|---|---|---|
| Integer Lattice | $1/6 = 0.1667$ | Literature minimum |
| Fibonacci | $0.2011 \pm 0.001$ | Confirms Zachary & Torquato (2009) |
| Silver | 0.2500 | Novel (later confirmed $= 1/4$ exactly; see §4) |
| Bronze | $0.2821 \pm 0.0006$ | Novel |
| Copper | 0.293 | Novel |
| Nickel | 0.310 | Novel |

### 3.4 Spreadability

Point patterns are embedded into two-phase media ($\phi_2 = 0.35$). The excess spreadability $E(t) = \mathcal{S}(\infty) - \mathcal{S}(t)$ yields $\alpha$ from the long-time slope $E(t) \sim t^{-(1+\alpha)/2}$.

| **Pattern** | $\alpha$ (measured) | $\alpha$ (expected) | Error |
|---|---|---|---|
| Fibonacci | 3.049 | 3 | 1.6% |
| Silver | 2.992 | 3 | 0.3% |
| Bronze | 2.987 | 3 | 0.4% |

Integer lattice confirmed $\alpha \to \infty$ (exponential decay). Spreadability was validated as a second, independent measure of $\alpha$.

## 4 Week 2

### 4.1 Uncorrelated Random Lattice and Stable Distributions

The uncorrelated random lattice (URL) displaces each site by $U(-a, a)$. The exact formula $\bar{\Lambda}(a) = (1 + a^2)/6$ was derived and validated to $< 0.3\%$ for all $a$. URL gives $\alpha = 2$ for all $a > 0$ (Class I). At $a = 1$ (cloaking limit), $\bar{\Lambda} = 1/3$ exactly — this turns out to be the limit of the metallic-mean sequence.

Symmetric $\alpha$-stable perturbations fill the $1 < \alpha < 2$ space stochastically (Class II/III for stability index $s < 1$).

### 4.2 Stealthy Hyperuniform Patterns

Stealthy configurations ($S(k) = 0$ for $|k| < K^*$) were analyzed:

- Collaborator-provided ensemble: ~4300 disordered-phase configurations per $\chi$ ($N = 2000$, $\rho = 1$, $\chi \in \{0.1, 0.2, 0.3\}$).
- 1D approximation derived: $\bar{\Lambda} \approx 1/(\pi^2\chi)$, validated below.

| $\chi$ | $\bar{\Lambda}$ (measured) | $\bar{\Lambda} = 1/(\pi^2\chi)$ | Excess |
|---|---|---|---|
| 0.10 | 1.021 | 1.013 | +0.8% |
| 0.20 | 0.526 | 0.507 | +3.7% |
| 0.30 | 0.357 | 0.338 | +5.6% |

**Two-phase discovery:** Generating stealthy configurations via collective-coordinate optimization from a *perturbed lattice* starting point gives $\bar{\Lambda} \approx 1/6$ — near the lattice minimum — while collaborator data (random starting positions) gives $\bar{\Lambda} \approx 1/(\pi^2\chi) \gg 1/6$. Both phases satisfy $S(k) = 0$ for $|k| < K^*$ but differ drastically in their spatial structure. This shows stealthy hyperuniformity does not uniquely determine a point pattern.

**Re-classification:** Stealthy patterns are not power-law suppressed; $\alpha$ is not meaningful. The correct descriptor is $K^* = 4\pi\rho\chi$.

### 4.3 New Substitution Chains

Three new chains were added to `substitution_tilings.py`:

| Chain | Rules | Matrix | Class | $\alpha$ |
|---|---|---|---|---|
| Period-Doubling | $S \to L$, $L \to LSS$ | $[[0,2],[1,1]]$ | II | 1.000 |
| 0222 Chain | $S \to LL$, $L \to SSLL$ | $[[0,2],[2,2]]$ | III | 0.639 |
| Bombieri-Taylor | $a \to aac$, $b \to ac$, $c \to b$ | $3 \times 3$ | I | 1.545 |

**Bombieri-Taylor** fills the $1 < \alpha < 2$ gap:

$$M_{BT} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda_1 = 2.247, \quad |\lambda_2| = 0.802, \quad \alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = \mathbf{1.545}, \quad \bar{\Lambda} = 0.377$$

This is the first known Class I quasicrystal with $1 < \alpha < 2$.

A new $\alpha = 2$ chain was also added (first deterministic substitution tiling at this value, previously only the stochastic URL occupied $\alpha = 2$):

$$a \to bc,\ b \to c,\ c \to abc \quad \Rightarrow \quad \lambda_1 \approx 2.148,\ |\lambda_{2,3}| \approx 0.682 \text{ (complex pair)},\ \alpha = 2.000,\ \bar{\Lambda} = 0.275.$$

### 4.4 Full Catalog Assembly and Junior Paper

All patterns were compiled into `catalog.json` via `research_catalog.py`. Infrastructure: `disordered_patterns.p` `stealthy_analysis.py`, `make_figures.py`.

The Junior Paper (`jp/jp_hyperuniformity.tex`) was drafted: §1 Introduction, §2 Theoretical Background, §3 Methods (5 subsections). 9 pages, biblatex+biber backend. Results sections (§4–§8) exist as a separate draft (`jp/jp_results_draft.tex`) pending organization.

## 5 Week 3

The following work was completed between the second and third meetings.

### 5.1 Bombieri-Taylor Deep Dive

**Script:** `bt_deep_dive.py` **Figure:** `fig_bt_deep.png`

1. BT has **32,517 Bragg peaks** in $k < 50$ vs. 2,637 for Fibonacci (12× denser).
2. **Rod overlap:** minimum inter-point spacing $= 1.000 < 2a = 1.002$; correcting $a$ changes the form factor.
3. **Slow convergence:** $|\lambda_2| = 0.802$ creates log-periodic oscillations with period $2\ln\theta_1 = 1.619$. At $t \in [10^5, 10^9]$: $\hat{\alpha} = 1.67$ (theory: 1.545). Converges extremely slowly from above.

**Conclusion:** Spreadability is unreliable for BT due to rod overlap and large $|\lambda_2|$. Use variance-based $\bar{\Lambda} = 0.377$ instead.

### 5.2 Finite-Size Scaling of $\alpha$

**Script:** `finite_size_scaling.py` **Figure:** `fig_finite_size_alpha_landscape.png`

Direct $S(k)$ power-law fitting fails for all substitution tilings (structure factor dominated by Bragg peaks). Variance-based metrics are reliable:

| **Pattern** | **Class** | **Metric** | $N = 1000$ | $N = 10^6$ |
|---|---|---|---|---|
| Bombieri-Taylor | I | $\bar{\Lambda}(N)$ | 0.3775 | 0.3783 (stable from $N = 2000$) |
| Fibonacci | I | $\bar{\Lambda}(N)$ | 0.1994 | 0.1982 (stable from $N = 1600$) |
| Period-Doubling | II | $C_{II}(N)$ | 0.1284 | 0.1233 (needs $N > 10^5$) |
| 0222 Chain | III | $\hat{\alpha}(N)$ | 1.046 | 0.776 (slow; theory = 0.639) |

The 0222 chain is slow because $|\lambda_2| = 1.236 > 1$ (not Pisot), causing large finite-size corrections even at $N = 1.5 \times 10^6$.

### 5.3 Systematic Search for New Substitution Chains

**Script:** `search_cubic_tilings.py` **Figure:** `fig_new_tilings.png`

Enumerated all 39,304 $3 \times 3$ non-negative integer matrices with row sums $\leq 4$. 4,882 pass the Pisot condition; 305 have nominal $\alpha \in (2, 3)$.

**Result:** All 305 give $\alpha = 2.000$ or $\alpha = 3.000$ exactly. No matrix fills $2 < \alpha < 3$.

**Proof:** For $3 \times 3$ unimodular with complex conjugate pair $\lambda_2 = \bar{\lambda}_3$:

$$|\det M| = 1 \Rightarrow \lambda_1 |\lambda_2|^2 = 1 \Rightarrow |\lambda_2| = \lambda_1^{-1/2} \Rightarrow \alpha = 2 \text{ exactly.}$$

**Extended to $4 \times 4$:** Search of 24M matrices (1.36M Pisot) confirms zero cases with $\alpha \in (2, 3)$ among unimodular matrices. Rank-$\geq$ 4 unimodular matrices with complex pairs give $\alpha < 2$ (additional real Pisot factor $\lambda_4$ forces $|\lambda_2|^2 < 1/\lambda_1$), so the conclusion holds for any rank.

### 5.4 Generalized Scaling Metrics for Classes II and III

**Script:** `generalized_metrics.py` **Figure:** `fig_generalized_metrics.png`

$$C_{II} = \lim_{R \to \infty} \frac{\sigma^2(R)}{\ln R} \qquad \text{(Class II)}$$

$$A_{III} = \lim_{R \to \infty} \frac{\sigma^2(R)}{R^{1-\alpha}} \qquad \text{(Class III)}$$

Results (curve-fit primary): Period-Doubling $C_{II} = 0.080$; 0222 Chain $A_{III} = 0.209$ (using $\alpha = 0.684$, fitted). Plateau cross-checks ($C_{II}^{\text{plat}} = 0.123$, $A_{III}^{\text{plat}} = 0.151$) are biased at finite $R$: plateau converges as $C + b/\ln R$ from above for Class II, and from below for Class III. Function `compute_generalized_metric()` added to `quasicrystal_variance.py`.

### 5.5 $\bar{\Lambda}$ Finite-$N$ Corrections

**Script:** `finite_size_lambda_bar.py` **Figure:** `fig_finite_size_lambda.png`

**Integer lattice:** The piecewise-cubic integral evaluates to $\bar{\Lambda} = 1/6$ exactly for all integer $R_{\max}$. No finite-$N$ correction exists; observed deviations ($< 0.3\%$) are Monte Carlo noise.

**Stealthy:** Ordered-phase stealthy (lattice start) gives $\bar{\Lambda} \approx 1/6$ at all $N$ and $\chi$, not $1/(\pi^2 \chi)$. This confirms the two-phase picture: both phases are stealthy but differ drastically in spatial structure.

### 5.6 Catalog Cleanup and Density Normalization

**Figure:** `fig_catalog_updated.png`

Stealthy patterns relabeled: $\alpha$ removed, $K^* = 4\pi\rho\chi$ added.

$\bar{\Lambda}$ is invariant under uniform density rescaling in $d = 1$: rescaling $x_i \to x_i/\rho$ leaves $\bar{\Lambda}$ unchanged ($< 0.2\%$ numerical error on Fibonacci). Direct comparison of $\bar{\Lambda}$ across all patterns with different densities is valid.

### 5.7 The $2 < \alpha < 3$ Gap: Non-Unimodular Extension

**Scripts:** `search_4x4_tilings.py`, `fig_nonunimodular_chain.py` **Figure:** `fig_nonunimodular.png`

The determinant constraint for all $2 \times 2$ Pisot matrices gives a unified formula:

$$|\lambda_2| = \frac{|\det M|}{\lambda_1} \implies \alpha = 3 - \frac{2\ln|\det M|}{\ln \lambda_1}.$$

For $|\det M| = 1$ (unimodular): $\alpha = 3$. For $1 < |\det M| < \sqrt{\lambda_1}$: $\alpha \in (2, 3)$.

A targeted search over $2 \times 2$ matrices with row sums $\leq 6$ found **28 matrices with** $\alpha \in (2, 3)$, all with $|\det M| = 2$. Minimum required row sum is 6 (one rule must produce at least 6 tiles).

**Smallest example:** $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ ($\det = -2$), rules $a \to b$, $b \to aabbbb$.

$$\lambda_1 = 2 + \sqrt{6} \approx 4.449, \quad |\lambda_2| = |2 - \sqrt{6}| \approx 0.449, \quad \alpha = 2.071.$$

Simulated at $N_{\text{iter}} = 9$: $N = 202{,}256$ points, $\rho = 0.2959$. $\sigma^2(R)$ shows **large log-periodic oscillations** over the range $[0.001, 0.87]$ with log-period $\ln \lambda_1 \approx 1.493$ in $\ln R$. Running average gives $\bar{\Lambda} \approx 0.650$ (rough estimate; large oscillation amplitude prevents precision measurement). This is the **first confirmed 1D substitution tiling with** $\alpha \in (2, 3)$.

### 5.8 High-Precision Silver $\bar{\Lambda}$ and the Metallic-Mean Limit

**Scripts:** `silver_precision_analysis.py`, `metallic_mean_convergence.py` **Figure:** `fig_metallic_converge`

**Silver:** High-precision run at $N = 3.88 \times 10^6$ ($3 \times 10^4$ windows, 1500 $R$ values) gives $\bar{\Lambda} = 0.2501 \pm 0.0004$, which is $0.3\sigma$ from $1/4$. The Silver chain has $\rho = 1/2$ exactly (mean spacing $= 2$). The result $\bar{\Lambda} = 1/4$ appears to be exact (novel; analytical proof not yet known).

**Metallic-mean limit:** $\bar{\Lambda}(\mu_n)$ was computed for $n = 1 \ldots 20$:

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| $\bar{\Lambda}$ | 0.2011 | 0.2500 | 0.2821 | 0.293 | 0.310 | 0.3177 | 0.3212 | 0.3249 | 0.3265 |
| $n$ | 15 | 20 | $\infty$ | | | | | | |
| $\bar{\Lambda}$ | 0.3289 | 0.3310 | 1/3 | | | | | | |

Convergence rate: $1/3 - \bar{\Lambda}(n) \approx 0.22/n^{1.43}$. The URL cloaked lattice ($a = 1$, $\bar{\Lambda} = 1/3$ exactly) is the limit of this family.

## 6 Complete Pattern Catalog

Patterns sorted by $\bar{\Lambda}$ within Class I.

| **Pattern** | **Class** | $\boldsymbol{\alpha}$ | $\bar{\boldsymbol{\Lambda}}$ |
|---|---|---|---|
| Integer Lattice | I | $\infty$ | $1/6 = 0.167$ |
| URL ($a = 0.1$) | I | 2.0 | 0.168 |
| URL ($a = 0.3$) | I | 2.0 | 0.181 |
| Gaussian $\sigma = 0.1$ | I | 2.0 | 0.187 |
| Fibonacci | I | 3.0 | 0.201 |
| URL ($a = 0.5$) | I | 2.0 | 0.208 |
| Silver | I | 3.0 | $\mathbf{1/4} = 0.250$ |
| Cubic $\alpha$=2 (new) | I | 2.0 | 0.275 |
| Bronze | I | 3.0 | 0.282 |
| Copper | I | 3.0 | 0.293 |
| Nickel | I | 3.0 | 0.310 |
| URL ($a = 1.0$) | I | 2.0 | 0.333 |
| Stealthy ($\chi = 0.3$) | I | —$^\dagger$ | 0.358 |
| Bombieri-Taylor | I | 1.545 | 0.377 |
| Stable $s = 1.7$ | I | 1.7 | 0.242 |
| Stable $s = 1.5$ | I | 1.5 | 0.303 |
| Stable $s = 1.3$ | I | 1.3 | 0.417 |
| Stealthy ($\chi = 0.2$) | I | —$^\dagger$ | 0.526 |
| Stealthy ($\chi = 0.1$) | I | —$^\dagger$ | 1.021 |
| Period-Doubling | II | 1.0 | div. ($C_{II} = 0.080$) |
| 0222 Chain | III | 0.639 | div. ($A_{III} = 0.209$) |
| Cauchy $\gamma = 0.1$ | II | 0.95 | div. |
| Stable $s = 0.7$ | III | 0.7 | div. |
| Stable $s = 0.5$ | III | 0.5 | div. |
| Stable $s = 0.3$ | III | 0.3 | div. |

$^\dagger$ Stealthy: $S(k) = 0$ by construction; $\alpha$ not applicable. Green rows: literature-confirmed benchmarks. Yellow: novel additions.

## 7 Figures and Scripts

### Figures

| Filename | Content |
|---|---|
| `fig1_poisson_benchmark.png` | Poisson variance validation |
| `fig2_bounded_variance.png` | $\sigma^2(R)$ for Fibonacci/Silver/Bronze |
| `fig3_hyperuniformity_test.png` | Normalized variance across classes |
| `fig9_stealthy_sk.png` | $S(k)$ for stealthy at $\chi = 0.1, 0.2, 0.3$ |
| `stealthy_Sk_overlay.png` | $S(k)$ overlay of all stealthy configs |
| `bombieri_taylor_analysis.png` | BT variance, $S(k)$, spreadability |
| `fig12_ranking.png` | Full $\bar{\Lambda}$ ranking, Class I |
| `fig_variance_II_III.png` | $\sigma^2(R)$ for Period-Doubling and 0222 |
| `fig_bt_deep.png` | BT oscillations, Bragg peak density, $E(t)$ |
| `fig_finite_size_alpha.png` | $\bar{\Lambda}(N)$, $C_{II}(N)$, $\hat{\alpha}(N)$ (portrait) |
| `fig_finite_size_alpha_landscape.png` | Same, landscape $1 \times 4$ for slides |
| `fig_new_tilings.png` | Cubic $\alpha$=2 chain variance and $S(k)$ |
| `fig_generalized_metrics.png` | Normalized variance plateaus (Class II/III) |
| `fig_finite_size_lambda.png` | Lattice $\bar{\Lambda}$ analytic vs. simulated |
| `fig_catalog_updated.png` | Updated $\bar{\Lambda}$ ranking with stealthy fixes |
| `fig_metallic_convergence.png` | $\bar{\Lambda}(\mu_n)$ for $n = 1 \ldots 20$ converging to 1/3 |
| `fig_4x4_alpha_dist.png` | $4 \times 4$ matrix search: $\alpha$ distribution |
| `fig_nonunimodular.png` | $\sigma^2(R)$ and running $\bar{\Lambda}$ for $\alpha = 2.071$ chain |

### Scripts

| Script | Purpose |
|---|---|
| `substitution_tilings.py` | All chains (metallic means, BT, cubic $\alpha$=2, non-unimodular) |
| `quasicrystal_variance.py` | $\sigma^2(R)$, $\bar{\Lambda}$, generalized metrics |
| `projection_method.py` | Cut-and-project Fibonacci chain |
| `two_phase_media.py` | Spectral density, spreadability |
| `disordered_patterns.py` | URL, stealthy generation, exact $\bar{\Lambda}$ for URL |
| `stealthy_analysis.py` | Collaborator stealthy ensemble analysis |
| `research_catalog.py` | Full pipeline $\rightarrow$ `catalog.json` |
| `make_figures.py` | Figures from catalog + stealthy results |
| `bt_deep_dive.py` | (Mtg 2–3) BT oscillations, $S(k)$ density, $E(t)$ |
| `finite_size_scaling.py` | (Mtg 2–3) $\bar{\Lambda}(N)$, $C_{II}(N)$, $\hat{\alpha}(N)$ |
| `search_cubic_tilings.py` | (Mtg 2–3) Exhaustive $3 \times 3$ matrix search |
| `generalized_metrics.py` | (Mtg 2–3) $C_{II}$, $A_{III}$; patches variance module |
| `finite_size_lambda_bar.py` | (Mtg 2–3) Lattice formula; stealthy two phases |
| `metallic_mean_convergence.py` | (Mtg 2–3) $\bar{\Lambda}(\mu_n)$ for $n = 1 \ldots 20$ |
| `search_4x4_tilings.py` | (Mtg 2–3) $4 \times 4$ matrix search (24M matrices) |
| `silver_precision_analysis.py` | (Mtg 2–3) High-precision Silver/Bronze $\bar{\Lambda}$ |
| `fig_nonunimodular_chain.py` | (Mtg 2–3) Figure for $\alpha = 2.071$ non-unimodular chain |

**Presentations and Written Work**

| Document | Location | Notes |
|---|---|---|
| Week 1 presentation | `results/week1/week1_presentation.tex` | Beamer, core metallic-mean results |
| Week 2 presentation | `results/week2/week2_presentation.tex` | Beamer, 12 slides |
| Week 3 presentation | `results/week3/week3_presentation.tex` | Beamer, 16 slides (rewritten 2026-03-24) |
| Junior Paper | `jp/jp_hyperuniformity.tex` | 9 pages, §1–§3 complete |
| JP results draft | `jp/jp_results_draft.tex` | §4–§8 + appendix, held for later |

All documents compile via MiKTeX (`C:/Users/minec/AppData/Local/Programs/MiKTeX/miktex/bin/x64/`).

## 8 Key Theoretical Results

1. $\alpha = 3$ **universality of metallic means.** For any $2\times 2$ metallic-mean matrix with $|\det M| = 1$: $|\lambda_2| = 1/\lambda_1 \Rightarrow \alpha = 3$ universally.

2. **Bombieri-Taylor fills $1 < \alpha < 2$.** The 3-letter cubic substitution $a \to aac, b \to ac, c \to b$ gives $\alpha = 1.545$, the first known 1D quasicrystal with $\alpha \in (1, 2)$.

3. **First deterministic $\alpha = 2$ substitution tiling.** The chain $c \to abc, a \to bc, b \to c$ gives $\alpha = 2.000$ exactly, $\bar{\Lambda} = 0.275$.

4. **$\bar{\Lambda}$ density invariance.** $\bar{\Lambda}$ is invariant under uniform density rescaling in $d = 1$; direct comparison across patterns with different densities is valid.

5. **Integer lattice has no finite-$N$ correction.** $\bar{\Lambda} = 1/6$ exactly for all integer $R_{\max}$ (piecewise cubic integral identity).

6. **Stealthy two-phase structure.** Both ordered and disordered stealthy phases satisfy $S(k) = 0$ for $|k| < K^*$, but differ drastically in $\bar{\Lambda}$ ($\approx 1/6$ vs. $1/(\pi^2\chi)$). Stealthy $\alpha$ is not a power-law exponent.

7. **$2 < \alpha < 3$ gap is inaccessible to unimodular matrices.** Exhaustive search over $3 \times 3$ (39,304) and $4 \times 4$ (24M) matrices confirms zero unimodular cases with $\alpha \in (2, 3)$. Rank-3 proof: $|\det M| = 1 \Rightarrow \alpha = 2$ exactly for complex pairs. Rank-$\geq 4$ proof: additional Pisot factor forces $\alpha < 2$.

8. **General $2\times 2$ formula.** For any $2 \times 2$ Pisot matrix: $\alpha = 3 - 2\ln|\det M|/\ln\lambda_1$. Gap $(2, 3)$ is accessible when $|\det M| \in (1, \sqrt{\lambda_1})$; all 28 known examples have $|\det M| = 2$.

9. **First 1D quasicrystal with $\alpha \in (2, 3)$.** $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ ($\det = -2$) gives $\alpha = 2.071$, $\bar{\Lambda} \approx 0.650$ (rough estimate). Large log-periodic oscillations in $\sigma^2(R)$, period $\ln\lambda_1 \approx 1.493$.

10. **Silver $\bar{\Lambda} = 1/4$ exact (numerical).** $N = 3.88 \times 10^6$, $3 \times 10^4$ windows: $\bar{\Lambda} = 0.2501 \pm 0.0004$ ($0.3\sigma$ from $1/4$). Analytical proof unknown.

11. **$\bar{\Lambda}(\mu_n) \to 1/3$ confirmed.** Metallic-mean sequence $n = 1 \ldots 20$ converges to $1/3$ at rate $0.22/n^{1.43}$. URL cloaked lattice ($a = 1$, $\bar{\Lambda} = 1/3$) is the exact limit.

## 9 Open Questions

1. $2 < \alpha < 3$ **gap:** ***RESOLVED*** **(mechanism and first example).** Non-unimodular $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ gives $\alpha = 2.071$. Sub-question: what is the minimal-complexity chain in $(2, 3)$? (Row sum $\geq 6$ is currently the minimum requirement.)

2. **Silver exact value:** ***RESOLVED numerically.*** $\bar{\Lambda} = 1/4$ at $0.3\sigma$. Analytical proof not yet known.

3. $\bar{\Lambda}$ **metallic-mean limit:** ***RESOLVED.*** $\bar{\Lambda}(\mu_n) \to 1/3$; rate $0.22/n^{1.43}$; URL cloaked lattice is the exact limit.

4. **Non-unimodular** $\bar{\Lambda}$**:** Is $\bar{\Lambda} = 0.650$ exactly $13/20$? Analytical proof unknown; large oscillation amplitude makes high-precision measurement difficult.

5. **Bronze closed form:** Is $\bar{\Lambda}(\text{Bronze}) = 0.2822$ a clean expression? Closest simple fraction: $11/39 = 0.28205$ ($0.18\sigma$, non-obvious).

6. **Stealthy two-phase literature:** Is the ordered/disordered bifurcation discussed in existing stealthy ground-state literature?

7. **2D generalization:** Does the eigenvalue formula $\alpha = 1 - 2\ln|\lambda_2|/\ln\lambda_1$ extend to 2D substitution tilings (Penrose, Ammann-Beenker)?